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General Knowledge
Indian Geography

STUDY NOTES

Major Crops of India

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India, with its diverse agro-climatic regions and rich agricultural history, supports the cultivation of a vast array of food and non-food crops. The country's agricultural landscape is molded by various soil types, climatic conditions, and long-standing traditional practices, leading to the growth of crops that are vital to the livelihoods of millions. This article provides an in-depth look at India's major crops, exploring their importance, cultivation methods, and regional distributions.

- **Kharif Crops (June-September):** Monsoon season, ideal for **rice, maize, cotton, pulses** (lentils, black gram, chickpeas, pigeon peas), **groundnut**, and **jute**.
- **Rabi Crops (October-March):** Winter season, with crops like **wheat, barley, pulses (gram, black gram)**, **mustard**, and **potato**.
- **Zaid Crop (March-April to June-July):** Short season with crops like **watermelon, muskmelon, cucumber,** and other **vegetables** like **tomatoes** and **okra**.

Commercial Classification

Food Crops	Rice, wheat, maize, millets — jowar, bajra, ragi; pulses — gram, tur (arhar), etc. **(cereals – grass-like plants with starchy edible seeds having high nutritional value)**
Cash Crops	Cotton, jute, sugarcane, tobacco, oilseeds, groundnut, linseed, sesamum, castor seed, rapeseed, mustard, etc.
Plantation Crops	Tea, coffee, coconut, areca nut, rubbe,r and spices — cardamom, chilies, ginger, turmeric, etc.
Horticulture	Vegetables — Onion, tomato, etc; and fruits —Apple, Orange, Mango, banana, citrus fruits, etc..

Rice

- **Temperature:** Between 22-32°C with high humidity.
- **Rainfall:** Around 150-300 cm.
- **Soil Type:** Deep clayey and loamy soil.
- **Top Rice Producing States: West Bengal > Punjab > Uttar Pradesh > Andhra Pradesh > Bihar.**
- It is the **staple food crop of the majority** of Indian people.
- India is the **second-largest producer of rice in the world after China.**
- In states like **Assam, West Bengal, and Odisha,** three crops of paddy are grown in a year. These are **Aus, Aman, and Boro.**
- **National Food Security Mission, Hybrid Rice Seed Production, and Rashtriya Krishi Vikas Yojana** are a few government initiatives to support rice cultivation.

Wheat

- **Temperature:** Between 10-15°C (Sowing time) and 21-26°C (Ripening & Harvesting) with bright sunlight.
- **Rainfall:** Around 75-100 cm.
- **Soil Type:** Well-drained fertile loamy and clayey loamy (Ganga-Satluj plains and black soil region of the Deccan)
- **Top Wheat Producing States: Uttar Pradesh > Punjab > Madhya Pradesh > Haryana > Rajasthan.**
- India is the **second-largest producer** after China.
- This is the **second most important cereal crop and the main food crop, in north and north-western** India.
- The success of the **Green Revolution** contributed to the growth of Rabi crops, especially wheat.
- **Macro Management Mode of Agriculture, National Food Security Missio,n and Rashtriya Krishi Vikas Yojana** are a few government initiatives to support wheat cultivation.

Millets (Nutri-Cereals)

- **Temperature:** Between 27-32°C
- **Rainfall:** Around 50-100 cm.
- **Soil Type:** Can be grown in inferior alluvial or loamy soil because they are less sensitive to soil deficiencies.
- **Jowar-** Rain-fed crop grown in moist areas with less or no irrigation.
- **Bajra-** Sandy soils and shallow black soil.
- **Ragi-** Red, black, sandy, loamy, and shallow black soils. (dry regions)
- **Top Millets Producing States: Rajasthan > Karnataka > Maharashtra > Madhya Pradesh > Uttar Pradesh**
- **Jowar:** Maharashtra > Karnataka > Madhya Pradesh > Tamil Nadu > Andhra Pradesh.
- **Bajra:** Rajasthan > Uttar Pradesh > Gujarat > Madhya Pradesh > Haryana.
- These are also known as **coarse grains,** which have **high nutritional value. Ragi** is very **rich in iron, calcium, other micronutrients**, and **roughage.**
- **Jowar** is the **third most important food crop with respect to area and production.**
- **National Agricultural Insurance Scheme, and Initiative for Nutritional Security through Intensive Millets Promotion** are examples of the government's efforts to support millet production.

Maize

- **Temperature:** Between 21-27°C
- **Rainfall:** High rainfall.
- **Soil Type:** Old alluvial soil.
- **Top Maize Producing States: Karnataka > Maharashtra > Madhya Pradesh > Tamil Nadu > Telangana**
- India is the **seventh largest producer** worldwide.
- It is used both as **food and fodder.**

- The use of modern inputs such as **High-Yielding Variety seeds, fertilizers, and irrigation** has contributed to the **increasing production** of maize.
- **Technology Mission on Maize** is one of the government's initiatives for Mazie.

Pulses

- **Temperature:** Between 20-27°C
- **Rainfall:** Around 25-60 cm.
- **Soil Type:** Sandy-loamy soil.
- **Top Pulses Producing States: Madhya Pradesh > Rajasthan > Maharashtra > Uttar Pradesh > Karnataka.**
- India is the **largest producer** as well as the **consumer** of pulses in the world.
- These are the **major source of protein** in a vegetarian diet.
- Major pulses grown in India are **tur (arhar), urad, moong, masur, peas and gram.**
- Being **leguminous crops,** all these crops except arhar help in restoring soil fertility by **fixing nitrogen from the air.** Therefore, these are mostly grown in rotation with other crops.
- **National Food Security Mission for Pulses, Pulses Development Scheme and Technological Mission on Pulses** are few of the government's plans to support pulses producti

Sugarcane

- **Temperature:** Between 21-27°C with a hot and humid climate.
- **Rainfall:** Around 75-100 cm.
- **Soil Type:** Deep rich loamy soil.
- **Top Sugarcane Producing States: Uttar Pradesh > Maharashtra > Karnataka > Tamil Nadu > Bihar.**
- India is the **second largest producer of sugarcane after Brazil.**
- It can be grown on all variety of soils ranging from sandy loam to clay loam given these soils should be well drained.
- It needs manual labour from sowing to harvesting.
- It is the main source of sugar, **gur (jaggery), khandsari and molasses.**
- **Scheme for Extending Financial Assistance to Sugar Undertakings (SEFASU)** and **National Policy on Biofuels** are two of the government initiatives to support sugarcane production and sugar industry.

Oil Seeds

- **Temperature:** Between 15-30°C
- **Rainfall:** Around 30-75 cm.
- **Soil Type:** Loam to clayey loam and well-drained sandy loams.
- **Top Oilseeds Producing States: Madhya Pradesh > Rajasthan > Gujarat > Maharashtra > Uttar Pradesh.**
- Main oil-seeds produced in India are **groundnut, mustard, coconut, sesamum (til), soyabean, castor seeds, cotton seeds, linseed and sunflower.**
- Most of these are edible and used as cooking mediums. However, some of these are also used as a **raw material in the production of soap, cosmetics and ointments.**
- **Yellow Revolution and Integrated Scheme on Oilseeds, Pulses, Oil Palm and Maize (ISOPOM)** are examples of government initiatives for oilseeds.

- **Groundnut is a kharif crop** and accounts for about **half of the major oilseeds produced** in the country.
- **Linseed and mustard are rabi** crops.
- **Sesamum is a kharif crop in north and rabi crop in south India.**
- **Castor seed is grown both as rabi and kharif crop.**

Horticulture Crops

- Horticulture is the **branch of agriculture** concerned with **cultivation, production and sale of fruits, vegetables, flowers, herbs, ornamental or exotic plants**
- India is the **second largest producer** of fruits and vegetables and it produces both **tropical and temperate fruits.**
- India produces about **13 percent of the world's vegetables.** It is an important producer of **peas, cauliflower, onions, cabbage, tomato, brinjal and potato.**
- **Golden Revolution, Mission for Integrated Development of Horticulture (MIDH) and Coordinated Horticulture Assessment and Management using geo-informatics (Project CHAMAN)** are government initiatives to help horticulture sector.

Tea

- **Temperature:** Between 20-30°C
- **Rainfall:** Around 150-300 cm.
- **Soil Type:** Deep and fertile well-drained soil, rich in humus and organic matter.
- **Top Tea Producing States: Assam > West Bengal > Tamil Nadu.**
- India is the **second largest producer** of tea.
- It was **introduced in the eastern hill slopes of India by the British.**
- Slopes of eastern hills have **humid climate and evenly distributed rainfall without water logging** which are optimal conditions for **terrace farming** of tea.
- Tea is a **labour intensive industry.** It requires abundant, cheap and skilled labour. Tea is processed within the tea garden to retain its freshness.
- **Tea Development and Promotion Scheme, Wage Compensation Scheme and Tea Boutiques** are few of the government schemes for tea.

Coffee

- **Temperature:** Between 15-28°C
- **Rainfall:** Around 150-250 cm.
- **Soil Type:** Well drained, deep friable loamy soil.
- **Top Coffee Producing States: Karnataka > Kerala > Tamil Nadu.**
- India is the **seventh largest producer.**
- **Coffee was initially brought from Yemen and introduced on the Baba Budan Hills.**
- **Hills with well-defined shade canopy, comprising evergreen leguminous trees** provide the optimal condition for coffee cultivation that is why it is mainly concentrated in the hilly regions.
- **Indian variety of coffee 'Arabica' is famous worldwide.**
- **Various Integrated Coffee Development Projects and schemes** have been launched by the government to support coffee production.

Rubber

- **Temperature:** Above 25°C with moist and humid climate.
- **Rainfall:** More than 200 cm.
- **Soil Type:** Rich well drained alluvial soil.
- **Top Rubber Producing States: Kerala > Tamil Nadu > Karnataka.**
- It is an **equatorial crop,** but under special conditions, it is **also grown in tropical and sub-tropical** areas.
- Rubber is an important **industrial raw material.**
- **Rubber Plantation Development Scheme and Rubber Group Planting Scheme** are examples of government led initiatives for rubber.

Cotton

- **Temperature:** Between 21-30°C
- **Rainfall:** Around 50-100cm.
- **Soil Type:** Well drained black cotton soil of Deccan Plateau.
- **Top Cotton Producing States: Gujarat > Maharashtra > Telangana > Andhra Pradesh > Rajasthan.**
- India is **believed to be the original home of the cotton plant.** Cotton is one of the main raw materials for cotton textile industry.
- Cotton needs **210 frost-free days and bright sun-shine for its growth.**
- It is a **kharif crop and requires 6 to 8 months to mature.**
- **Silver Fibre Revolution and Technology Mission on Cotton** are the government initiatives for increasing cotton production in India.
- Cotton has been genetically modified and BT Cotton to **fight environmental stress and pest attacks.**

Jute

- **Temperature:** Between 25-35°C
- **Rainfall:** Around 150-250 cm
- **Soil Type:** Well drained alluvial soil
- **Top Jute Producing States: West Bengal > Bihar > Assam > Andhra Pradesh > Odisha.**
- It is **mainly concentrated in eastern India** because of the **rich alluvial soil of Ganga-Brahmaputra delta.**
- India is the **largest producer of jute.**
- It is known as the **golden fibre.**
- It is used in making **gunny bags, mats, ropes, yarn, carpets and other artefacts.**
- Due to its **high cost, it is losing market to synthetic fibres and packing materials, particularly nylon.**
- **Golden Fibre Revolution and Technology Mission on Jute and Mesta** are two of the government initiatives to boost jute production in India.

Challenges in Indian Agriculture

India's agricultural sector faces several challenges, including:

- **Agricultural Stagnation**: India's crop yields are often lower than global standards, with many farmers expressing the desire to exit farming due to low profits.
- **Small Landholdings**: With nearly 86% of farmers owning less than two hectares, small and marginal landholdings limit agricultural productivity and sustainability.
- **Hunger and Poverty**: Despite high food grain production, hunger and malnutrition remain prevalent, with India ranking 101st out of 116 countries on the Global Hunger Index in 2021.
- **Food Wastage**: Around 67 million tonnes of food, worth approximately Rs 92,000 crores, is wasted annually, with significant spoilage occurring in storage facilities.
- **Climate Change**: Rising temperatures impact crop yields and introduce new pest challenges, particularly affecting crops like wheat and rice.
- **Dryland Agriculture**: Accounting for 42% of total food grains, dryland areas are vulnerable to climate change and require improved soil and moisture conservation methods.
- **Agro-Infrastructure**: A lack of adequate infrastructure—such as irrigation, storage for perishables, and cold chains—limits agricultural efficiency and market accessibility.

Risks in Agriculture and Suggested Measures

To manage agricultural risks, suggested measures include:

Type of Risk	Causes	Severity	Measures
Production Risks	Pest attacks, crop diseases	Low productivity	Disease-resistant seeds, fair markets for inputs
Weather/Disaster Risks	Drought, floods, unseasonal rain	Production loss	Irrigation coverage, water management
Price Risks	Low prices	Profiteering by middlemen	Infrastructure development, market accessibility
Credit Risks	Reliance on informal loans	Farmer indebtedness	Increase formal credit availability (e.g., Kisan Credit Card)
Market Risks	Changing demand	Loss of market share	Long-term purchase contracts, contract farming
Policy Risks	Uncertain policies	Impact on market and production	Stable trade and policy changes

Conclusion

Despite these challenges, India remains a leading global agricultural producer, thanks to its diverse agroecological regions and resilient farming practices. To enhance productivity, improve market access, and support climate-resilient farming, strategic initiatives are essential for India to continue its role as a global agricultural leader while ensuring national food security.